Girls Build Bots — Cadette STEM Career Exploration

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** STEM Career Exploration **Format:** One 2-hour session · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. This badge is the one where **the team itself is the main resource** — every mentor is a live example of a STEM path in progress. A robot demo is a strong bonus but not required.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette STEM Career Exploration badge. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The big idea

Cadettes are close enough to high school that career exploration gets practical: what do STEM jobs *actually involve* day to day, what's the pathway (classes, clubs, degrees), and what can a middle schooler do *this year* that counts? The team's honest answer — "we're on that pathway right now, here's what it looks like" — is the badge's superpower.

Vocabulary: career pathway, day-in-the-life, internship, portfolio, mentor, informational interview.

STEM Career Exploration (2 hours)

Materials

- "Day in the life" career cards, each with the honest version: a software engineer (mostly reading and fixing code, meetings, teamwork), a biomedical engineer, a data scientist, a robotics technician, a UX designer, an environmental engineer — real tasks, typical pathway, and one myth busted per card
- Informational-interview question sheets; pathway-map worksheets (now → high school → after)
- Team members prepped for panel + interviews; optional robot for demos

Activity 1 — Myth vs. Day-in-the-Life (20 min)

Deal the career cards. Teams read and report the surprise: what does this job *actually* involve vs. the TV version? (Software engineers spend more time reading code than writing it; scientists spend a lot of time writing.) Point: choosing a career means knowing its Tuesdays, not its highlight reel.

Activity 2 — The Panel, Cadette Edition (25 min)

Team members present 3-minute versions of their story with the practical layer Cadettes want: what classes they chose and why, how they joined the team, what they're aiming for after high school, and one honest struggle. Center the young women on the team. Then open Q&A — Cadettes ask sharper questions than anyone expects; let them.

Activity 3 — Informational Interviews (35 min)

The badge's core skill, done for real: in pairs, Cadettes conduct 8-minute **informational interviews** with team members using the question sheet (What does your role involve? What surprised you? What would you tell someone my age? What's the next step on your path?), taking notes, then rotating once. Interviewing a person about their path is a career skill they'll use for decades — say so.

Activity 4 — Pathway Map + Commitment (25 min) + Wrap (15 min)

Each Cadette builds her **pathway map**: a career direction that interests her → what she can do *now* (a club, a free course, FLL/FTC, a science fair) → high school moves (classes, teams) → after. The map is a draft, not a contract — paths change, and the panel just proved it. Closing round: each Cadette shares her now-step and *when* she'll take it. Optional robot demo finale — hands on the controls.

> **Robot moment (optional):** Let Cadettes drive the robot, then land the line: "Every person who built this started with a step exactly the size of the one on your map."

Completion Checklist — STEM Career Exploration

By the end of the session, to earn the badge, each Scout can: - [ ] Describe the realistic day-to-day of at least two STEM careers - [ ] Conduct an informational interview and record what she learned - [ ] Map a pathway from a career interest back to actions she can take now - [ ] Commit to one concrete next step with a timeframe

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Mentor Guide (for the robotics team)

- **You are the curriculum.** Prep the practical layer: your class choices, how you joined, your next step. Cadettes want the map, not just the inspiration.
- **Be honest about Tuesdays.** The unglamorous parts of your role build more trust than the highlight reel — and trust is what makes the pathway feel reachable.
- **Center the young women on your team** in the panel and interviews — for Cadettes weighing whether STEM is "for them," representation is the argument.
- **Take the interviews seriously.** Give real answers, let them take notes, treat it as practice for the real thing — because it is.
- **Every Cadette leaves with a dated step.** "Someday" doesn't earn the badge; "FLL signup by Friday" does.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Cadette STEM Career Exploration badge; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.